PAPER CODE NO.
COMP337/527

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SECOND SEMESTER EXAMINATIONS 2024/25

Class Test

TIME ALLOWED : One hour forty minutes

NAME OF CANDIDATE . . . .
STUDENT ID . . . .

THIS PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

1. Let $\overline{X}$ be a vector $[42, -81, -10, 21, 42, -1]$. What will be the value of $\overline{X_6}$ after 0/1 scaling. **5 marks**

2. In a one-vs-rest strategy, assume the following decision scores for a test sample:

   - Class 1: 0.6
   - Class 2: 0.4
   - Class 3: 0.9

   Which class will the classifier assign to the sample?

   5 marks

3. **Y**ou are optimizing a quadratic function p^2 using gradient descent with an initial point 1 and a learning rate of 0.1. What will be the value of p after one iteration of gradient descent? **5 marks**

4. You flip a fair coin 3 times. What is the probability of getting exactly 2 heads? The probability of head or tail in a single flip is $1/2$. **5 marks**

5. Compute the derivative of the function $x^3 cos^2 x$ **5 marks**

6. Find the eigen value of the following matrix.
   $\begin{pmatrix} -6 & 3 \\ 4 & 5 \end{pmatrix}$ **5 marks**

7. Find the rank of the following matrix. $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ **5 marks**

8. Find the derivative of the function $x^3 log(x)$

9. Find A+B for the matrices
   A= $\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$ and B= $\begin{pmatrix} 4 & 2 \\ 0 & 7 \end{pmatrix}$ **5 marks**

10. Find the Manhattan distance between the vectors $\overline{X} = [1, 2, -5]$ and $\overline{Y} = [6, 3, -5]$ **5 marks**